

LECTURE 1

Foundations, Paradigms, Defence Applications & Data Pitfalls

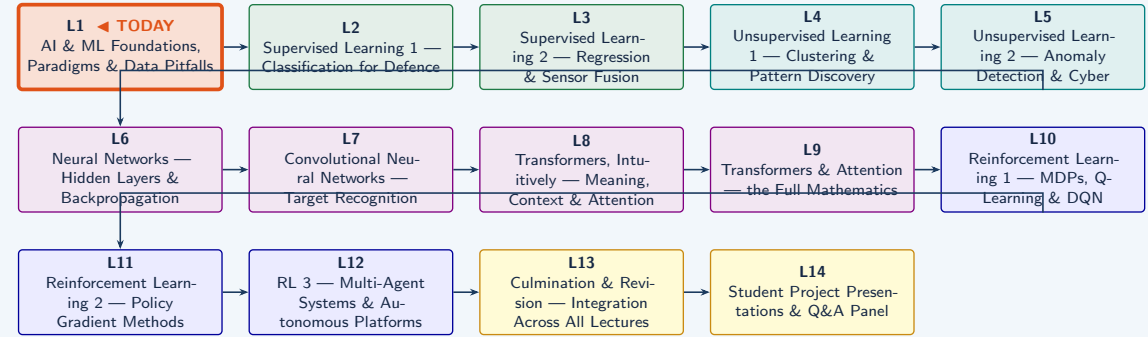
AI Foundations • ML Paradigms • Defence & Wargaming Applications • Data Pitfalls

AK DHAMIJA, Researcher & Instructor

AI & Machine Learning

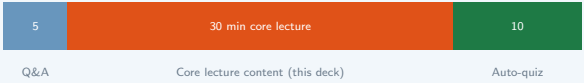
dhamija.ak@gmail.com • 9811453163

Unit 4 Lecture Schedule: 14 Lectures — one story from data to autonomous decisions



Reading the map: L1 sets the vocabulary and the data-quality lens; **L2–L3** teach prediction with labels; **L4–L5** teach discovery without labels; **L6–L9** open the neural-network black box (hidden layers, backpropagation, CNNs, transformers); **L10–L12** teach decision-making by interaction; **L13–L14** consolidate and assess.

Lecture structure (45 min)



- **5 min** — Q&A on previous session
- **30 min** — core lecture content (this deck)
- **10 min** — auto-quiz (15 questions × 40 s)
- Format: 4-pane layout, equations, defence examples

DATA PITFALLS thread woven throughout — a critical lens for every AI/ML practitioner in Defence Research. Based on AK Dhamija (SlideShare: Goody Research) plus established ML literature.

Parts of each lecture

Part 1: lecture slides **Part 2:** auto-quiz (40 s/Q)
Part 3: quiz answers + assignment solutions **Part 4:** instructor manual

Learning objectives

By the end of this lecture you can...

1. Define AI, ML and core paradigms precisely
2. Trace evolution: Classical AI → GenAI
3. Distinguish the 5 ML learning paradigms
4. Describe Deep Learning key architectures
5. Explain LLMs and RAG in defence context
6. Map AI/ML to Defence & Wargaming use cases
7. Spot flaws in data analysis & interpretation (Simpson, H_0, correlation vs causation)
8. Identify and avoid 6 critical data pitfalls

Anchor: all numerical examples & case studies are anchored in Defence R&D and Wargaming Simulation contexts.

Syllabus — Unit 4

AI & ML Paradigms | Supervised Learning for Defence | Unsupervised Learning & Clustering | RL for Autonomous Systems

Before Any Algorithm: Data Can Lie — Six Famous Failures, Six Disciplines

Every one of these errors has a twin in defence analysis and wargaming

AERONAUTICS — 1943

Engineers wanted to armour where returning bombers had bullet holes. Wald: armour where they had *none*.
Error: survivorship bias.
Wargame twin: learning only from won engagements.

ENGINEERING — 1986

Challenger O-ring analysis looked only at flights *with* damage; the 17 damage-free warm-weather flights were left out — hiding the temperature trend.
Error: missing / selected data.
Twin: dropping "no-event" sorties.

PSYCHOLOGY — 1960s

Israeli Air Force instructors: praise after good landings was followed by worse ones; criticism by better ones. Truth: regression to the mean.
Error: attributing chance to cause.
Twin: "retraining fixed the worst units".

BIOLOGY / MEDICINE — 1990s

A pneumonia-risk model learned that asthma *lowers* death risk — because asthmatics were sent straight to intensive care.
Error: confounding by treatment.
Twin: "units under air cover take more losses".

STATISTICS — 1973 / 1986

Berkeley admissions and kidney-stone trials: a treatment or group that wins in *every* subgroup loses in the total.
Error: Simpson's paradox.
Twin: "AI-equipped units win less".

DEFENCE — 1988 / 2003

USS Vincennes downed an airliner the crew believed was attacking; Patriot batteries engaged two friendly aircraft. Data and automation were trusted over doubt.
Error: expectation & automation bias.
Twin: rubber-stamping AI alerts.

The point of this section: algorithms do not fix these errors — they *automate* and *scale* them. Before we learn a single model, we learn how data and interpretation fail.

Simpson's Paradox (I) — Never Add or Average Percentages: Use the Counts

Kidney-stone treatments (Charig et al., BMJ, 1986): A = open surgery, B = percutaneous nephrolithotomy

Stone size	Treatment A	Treatment B
Small stones	**81 / 87 = 93%**	234 / 270 = 87%
Large stones	**192 / 263 = 73%**	55 / 80 = 69%
Combined	273 / 350 = **78%**	289 / 350 = **83%**

Cells show successes / patients = success rate. A wins in BOTH subgroups yet loses overall.

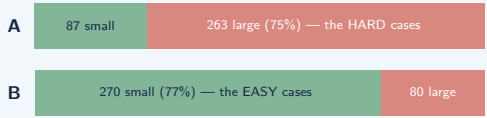
Tempting mistake: adding percentages ($93+73 = 166$ vs $87+69 = 156$) or averaging them (83% vs 78%) — meaningless: it ignores group sizes.

COMBINED RATE = WEIGHTED AVERAGE BY GROUP SIZE

$$\text{A: } \frac{87(0.93)+263(0.73)}{350} = \frac{81+192}{350} = 78\%$$

$$\text{B: } \frac{270(0.87)+80(0.69)}{350} = \frac{234+55}{350} = 83\%$$

Why: case mix differs between treatments



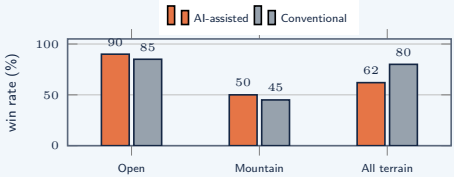
- Large stones are harder to treat *and* were mostly given treatment A
- So A's total is dragged down by case mix, not by worse treatment
- **Confounder:** stone size (severity) drives both the choice of treatment and the outcome

Rule: always compute totals from raw counts, show the counts beside every percentage, and stratify by the variables that drive both assignment and outcome.

Simpson's Paradox (II) — The Wargaming Twin: "AI-Assisted Units Perform Worse"?

Exercise records: engagements won / fought, split by terrain (illustrative teaching data)

Terrain	AI-assisted units	Conventional units
Open / easy	**81 / 90 = 90%**	238 / 280 = 85%
Mountain / hard	**105 / 210 = 50%**	18 / 40 = 45%
All terrain	186 / 300 = **62%**	256 / 320 = **80%**



Headline a staff officer might write: "AI-assisted units win only 62% vs 80% — withdraw the AI tool."

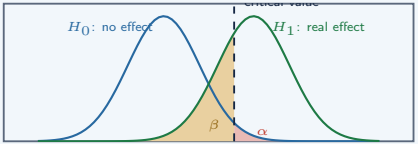
The truth: AI-assisted units win MORE in every terrain (+5 points in both). They were deliberately sent to the hard mountain sectors: 210 of their 300 engagements (70%) vs 40 of 320 (12.5%) for conventional units.

Same pattern, real world: UC Berkeley 1973

Overall admission: men 44%, women 35%. Department by department there was no bias against women — women applied more to the most competitive departments (Bickel, Hammel & O'Connell, *Science*, 1975).

Ask every time: what decided which units, patients or sorties ended up in each group? That variable must be stratified or controlled.

The Null Hypothesis & Statistical Significance — What a p-value Says



	H_0 true (no effect)	H_0 false (real effect)
Reject H_0	Type I error (α): false alarm	Correct (power $= 1 - \beta$)
Keep H_0	Correct	Type II error (β): missed effect

p-value = probability of data at least this extreme *if H_0 were true.* It is NOT the probability that H_0 is true, and NOT the size or importance of the effect.

Defence example: radar-absorbent coating trial (illustrative)

H_0: the new coating does not change detection probability. 40 sorties each: baseline detected **26/40 (65%)**, new coating **18/40 (45%)**.

TWO-PROPORTION Z-TEST

$$\hat{p} = \frac{44}{80} = 0.55$$

$$SE = \sqrt{0.55(0.45)\left(\tfrac{1}{40} + \tfrac{1}{40}\right)} = 0.111$$

$$z = \frac{0.65 - 0.45}{0.111} = 1.80 \;\Rightarrow\; p \approx 0.072 \quad (> 0.05)$$

Wrong conclusion: "the coating doesn't work". A 20-point drop is large; the trial was under-powered. **Absence of evidence $\neq$ evidence of absence.** Power calculation: detecting 65%→45% with 80% power at $\alpha = 0.05$ needs $\approx$ **93** sorties per arm, not 40.

"Significant" is not "important"

Huge sample, trivial effect (illustrative)

1,000,000 simulated engagements per variant: hit rate 50.0% vs 50.2%. $SE = \sqrt{0.25 \cdot 2/10^6} = 0.0007$, $z = 0.002/0.0007 \approx 2.8$, $p \approx 0.005$: "highly significant" — yet 0.2 points will never change a tactical decision.

Report what matters

Effect size (difference, ratio, Cohen's d) with a **95% confidence interval**, and the cost/benefit of acting on it. A wide CI that includes zero *and* large effects means "we don't know yet" — collect more data.

Psychology's warning: the replication crisis

When 100 published psychology studies were repeated, only about 36% gave a significant result again (Open Science Collaboration, *Science*, 2015). Small samples + flexible analysis + publication of "significant" results only.

Run enough tests and something will be "significant"



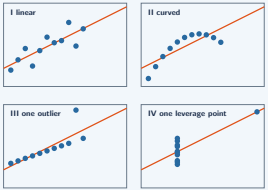
FAMILY-WISE ERROR AND BONFERRONI CORRECTION

$$P(\geq 1 \text{ false positive}) = 1 - (1 - \alpha)^m$$

Bonferroni: $\alpha_{\text{per test}} = \alpha/m$

Wargame trap: testing 20 tactics, sectors or time windows and reporting the one with $p < 0.05$ is p-hacking. Pre-register the hypothesis, correct for multiple tests, and confirm on fresh exercise data.

Anscombe's quartet (statistics, 1973): four datasets, identical summary statistics



All four: mean $x = 9$, mean $y = 7.50$, variances equal, $r = 0.816$, fitted line $y = 3.00 + 0.500x$.

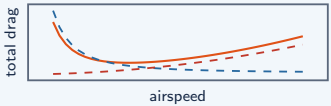
Rule: *plot* the data before trusting r, a regression line or an AI feature-importance score.

PEARSON CORRELATION

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \in [-1, 1]$$

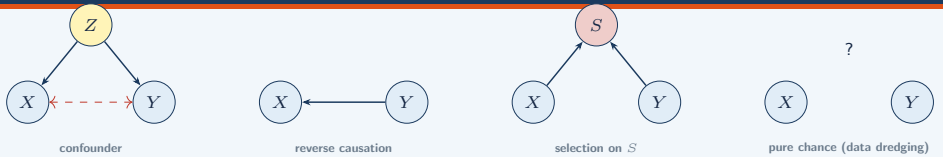
- r measures only **linear** association; it is driven by outliers and leverage points
- r^2 = share of variance explained by the straight line

Aeronautics: $r \approx 0$, yet perfect dependence



Induced drag falls and parasite drag rises with speed: the U-shaped total has near-zero linear r over the envelope.

Correlation $\neq$ Causation — Spurious Correlation and How It Arises



Statistics / biology

Storks and babies: across 17 European countries, stork pairs correlate with birth rate ($r \approx 0.62$, Matthews 2000). Confounder: land area and rural population. **Ice-cream sales and drownings:** both driven by summer heat.

Engineering — reverse causation

Vehicles with the most maintenance hours have the most breakdowns. Maintenance does not cause failure: failing vehicles *attract* maintenance.

Defence & wargaming

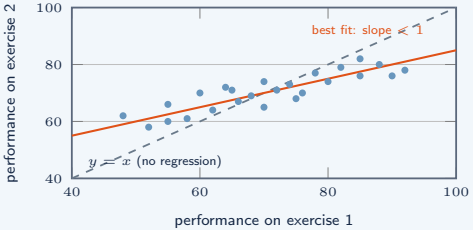
Confounder: "catering-quality score" vs exercise win rate, $r = 0.71$; controlling for unit budget/readiness, $r_{\text{partial}} = 0.09$ ($p = 0.42$). **Selection:** analysing only engagements that were reported upward.

Data dredging

Screen 1,000 candidate indicators with no real effect at $\alpha = 0.05$: expect $\approx$**50** "significant" correlations by chance alone. AI feature-mining does exactly this at scale.

To claim causation: a controlled experiment (randomised trial, A/B wargame), or a causal design that blocks confounders — plus a plausible mechanism.

Regression to the Mean — When Chance Masquerades as Cause



Illustrative data. Whenever luck contributes to a score, extreme scorers move toward the average next time — with no intervention at all.

Psychology + aeronautics: Israeli Air Force flight school

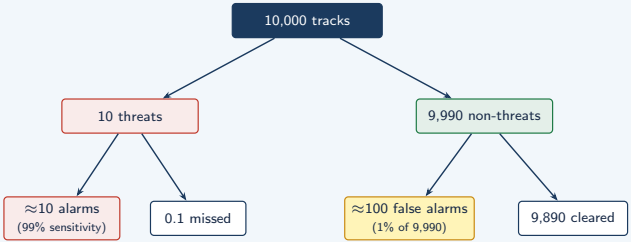
Instructors observed that praise after an excellent landing was usually followed by a worse one, and criticism after a bad landing by a better one — and concluded that punishment works. Kahneman showed it was regression to the mean: an exceptional landing is partly luck.

Biology: where the word comes from

Galton (1886): children of very tall parents are tall, but less extreme — "regression towards mediocrity". The statistical term "regression" (Lecture 3) is named after this effect.

Defence trap: "the five worst-scoring battalions improved after the new AI training aid". They would have improved anyway. **Fix:** compare against a control group of equally poor units that did not get the intervention.

10,000 air tracks; 10 are real threats (base rate 0.1%)



BAYES' THEOREM: POSITIVE PREDICTIVE VALUE

$$P(\text{threat} \mid \text{alarm}) = \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.01 \times 0.999} \approx \frac{10}{110} \approx \mathbf{9\%}$$

Biology / medicine

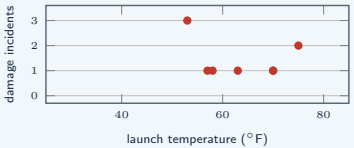
A screening test with 99% sensitivity and 99% specificity for a disease with 1% prevalence: only **50%** of positives are truly ill. Doctors routinely over-estimate this figure.

Defence consequence: 10 of every 11 IADS alarms are false. Operators learn to distrust the system (alarm fatigue) — or engage everything. Either way the "99% accuracy" brochure figure misleads.

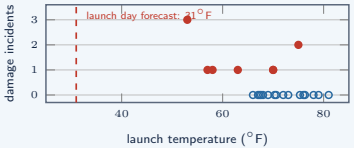
Always ask: what is the base rate in the *deployment* stream? Report precision (PPV) at that base rate, and design a second-stage check (human or sensor) for alarms.

Engineering Lesson: Challenger (1986) — The Data You Leave Out

What was examined: only flights with O-ring damage



All flights: damage-free flights included



Approximate reconstruction of the pre-launch flight record (cf. Dalal, Fowlkes & Hoadley, *JASA*, 1989).

What happened

On the eve of launch, engineers and managers reviewed O-ring damage against temperature using mainly the flights that *had* damage. The picture looked flat. Every flight below 66°F had damage. Challenger launched at about 31°F — far colder than any previous flight — and was lost with its crew of seven.

Two compounding errors: (1) excluding "no-event" cases removes the contrast that reveals a trend; (2) extrapolating far outside the observed range.

Defence twin: reliability or vulnerability analyses that keep only sorties with incidents, and AI models asked to predict in conditions (temperature, terrain, EW intensity) never seen in training.

AI Interpretation Errors (I) — Shortcut Learning: Right Answer, Wrong Reason

PSYCHOLOGY — "Clever Hans" (1907)
A horse appeared to do arithmetic. Pfungst showed it read tiny cues in the questioner's posture. The original "shortcut learner".
AI lesson: accuracy alone cannot tell you *what* was learned.

COMPUTER VISION — husky vs wolf
A classifier separated wolves from huskies using the *snow* in the background (Ribeiro et al., 2016, LIME). Explanation tools exposed it.
Defence twin: target IDs keyed on terrain, not vehicle.

MEDICINE — pneumonia risk
The model rated asthma patients as *lower* risk: in the data they were rushed to intensive care and survived (Caruana et al., 2015).
Defence twin: "high-threat sectors show low losses" because they got the most air cover.

FOLKLORE — the "tank detector"
An often-told (probably apocryphal) story: a network "detected tanks" by learning that tank photos were taken on cloudy days.
Lesson: the risk is real even if the tale isn't.

ENGINEERING — adversarial stickers
A few stickers on a stop sign made a vision model read it as a speed-limit sign (Eykholt et al., 2018).
Defence twin: camouflage patterns or decoys designed against our classifiers.

WARGAMING — metadata leakage
A model "predicts" night attacks from timestamp fields rather than thermal signatures — perfect in the lab, useless when OPFOR changes its schedule.
Lesson: audit inputs for leakage.

Defences: explanation tools (SHAP, LIME, saliency maps); tests on deliberately shifted data; red-team adversarial inputs; remove leaky metadata before training.

Automation bias — Patriot, 2003

During the 2003 Iraq campaign, Patriot batteries shot down a British Tornado and a US Navy F/A-18, killing three aircrew. Contributing factors included misclassification of friendly aircraft and operating procedures in which crews accepted automated engagement cues under time pressure.

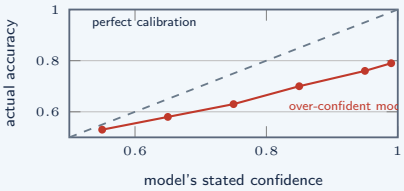
Expectation (scenario fulfilment) — USS Vincennes, 1988

Crew under combat stress perceived Iran Air 655 as a descending attacking fighter, although the data showed a climbing airliner. 290 people died. Psychology: people see what they expect.

Confirmation bias in labels

Analysts label ambiguous tracks to fit the current threat picture; the model then learns the analysts' bias and reflects it back with apparent objectivity.

Confidence $\neq$ correctness: calibration



Illustrative: a "99% confident" output may be right only ∼80% of the time, worse under distribution shift.

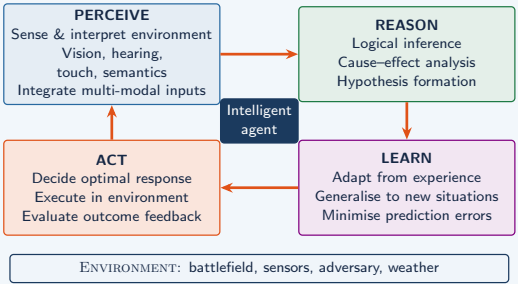
Countermeasures: calibrate scores; display uncertainty, not just labels; train operators on known failure modes; require positive human confirmation for lethal action; log decisions for after-action review.

The Analyst's Oath — Ten Rules Before Trusting Any Data or AI Output

1. **Count, don't average percentages.** Totals come from raw numbers (Simpson).
2. **Ask what is missing.** Survivors, damage-free flights, unreported engagements.
3. **Stratify by what drove assignment.** Terrain, severity, budget.
4. **State H_0 and compute power before testing.** Don't under-test.
5. **Report effect sizes with confidence intervals,** not just p.
6. **Correct for multiple tests;** confirm on fresh data.
7. **Plot the data** — Anscombe; one number hides the shape.
8. **Correlation is a question, not an answer.** Seek mechanism + experiment.
9. **Check the base rate** before trusting any alarm or "99% accuracy".
10. **Test on the deployment domain and keep a human accountable.**

An AI system is only as trustworthy as the data it learned from and the person interpreting it. Every lecture that follows applies these ten rules.

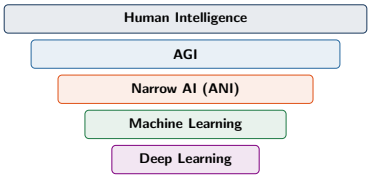
Intelligence: Perceive → Reason → Learn → Act — a closed loop



Warning: current AI systems are **Narrow AI** — excellent at specific tasks, but with no common sense or cross-domain reasoning.

WARGAMING INTELLIGENCE: a Blue Force AI agent must **PERCEIVE** the battle map (sensor fusion), **REASON** about OPFOR intent, **LEARN** from prior engagements, and **ACT** to maximise mission success — mirroring all four cognitive pillars. This is the entire motivation for AI in warfare.

Where is today's AI on the intelligence spectrum?



Human Intelligence ⊃ AGI ⊃ Narrow AI ⊃ ML ⊃ Deep Learning

Defining AI — From Turing (1950)

John McCarthy (1956)

"The science and engineering of making intelligent machines, especially intelligent computer programs capable of performing tasks that typically require human intelligence."

Russell & Norvig (AIMA, 2020)

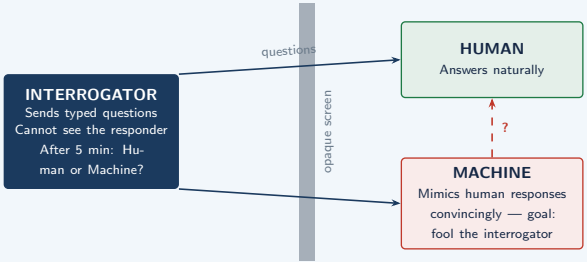
"AI is the study of agents that receive percepts from the environment and perform actions. Each such agent implements a function that maps percept sequences to actions."

AGENT FUNCTION (RUSSELL & NORVIG)

$$f : \text{percept sequences} \longrightarrow \text{actions}$$

Key point: LLMs (GPT-4, Claude) reliably pass the Turing Test in text, yet "passing" $\neq$ "truly intelligent". In defence AI what matters is **task performance**, NOT the appearance of intelligence.

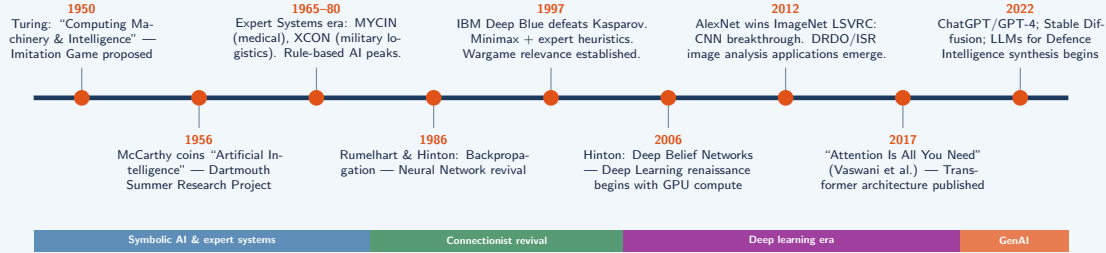
The Turing Test (1950): still relevant?



Defence angle

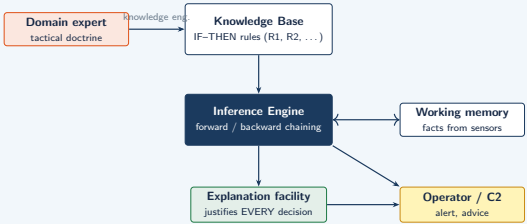
Can an AI system blend into legitimate SIGINT/COMMS traffic (EW context)? Deep-fake voice in strategic deception? Turing-style indistinguishability is an **adversary capability** we must detect, not only a milestone to admire.

AI Timeline: From Turing to Transformers to Defence Deployments



Pattern: each wave was driven by a bottleneck being lifted — rules (1960s) → data & backprop (1980s) → GPUs & ImageNet (2012) → attention & scale (2017+). Periods of over-promise were followed by "AI winters"; defence adoption has always lagged civil capability by 5–10 years.

Expert Systems: Rule-Based AI — The First Defence AI Wave



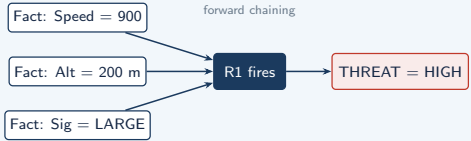
Advantages & inference modes

✓ Fully transparent & auditable ✓ No training data required ✓ Deterministic, repeatable ✓ Expert knowledge encoded directly.
Forward chaining: facts → conclusion. **Backward:** goal → supporting facts.

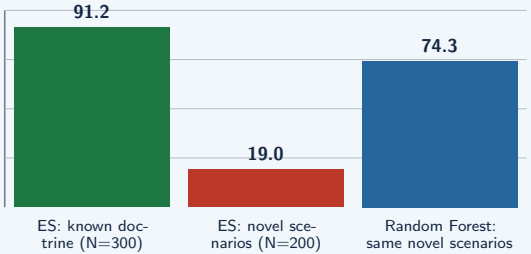
Defence example: IADS threat assessment

```
R1: IF RadarSig = ‘‘LARGE’’ AND Speed > 800 km/h
    AND Altitude < 300 m
  THEN ThreatLevel = HIGH, Alert = LAUNCH
R2: IF IFF_Code = ‘‘FRIEND’’ AND Course =
‘‘RETURNING’’
  THEN ThreatLevel = NIL, Allow_Landing
R3: IF EW_Jamming = TRUE AND Dir = ‘‘NORTH’’
  THEN CyberAlert, Notify_HQ
```

Real systems: Cold War SAGE & BMEWS expert systems had **100,000+** encoded rules — all hand-written.



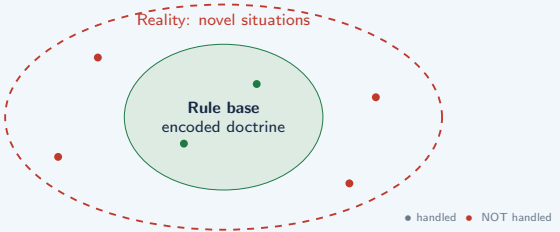
Artillery Fire Assignment ES: known vs novel scenarios



Percent correct on Artillery Fire Assignment decisions

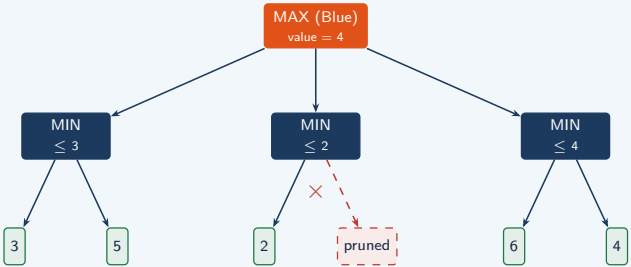
- Rules encoded: **500** (200 expert-months); response $<$ **0.08 s** per query
- Known doctrine: **91.2%** correct; novel/adapted: only **19.0%**
- Random Forest on the same novel scenarios: **74.3%**

Why the cliff? The knowledge bottleneck



KEY INSIGHT: expert systems excel in CLOSED, WELL-DEFINED domains but fail catastrophically when reality diverges from the rule base. The knowledge bottleneck: encoding 200 years of tactical doctrine into IF–THEN rules is impossible — and **war is inherently novel**.

Game Trees, A* Search & the Limits of Classical AI in Warfare



- Leaf = evaluation of the final board; MIN picks the smallest child, MAX the largest.
- After the left branch gives 3, the middle branch already shows $2 < 3$: MIN's remaining leaf can never change MAX's choice $\Rightarrow$ α–β **cut-off**.

Game-tree search (minimax)

Two-player zero-sum game (chess, wargame). **MAX** (Blue Force) maximises own score; **MIN** (Red Force) minimises Blue's score. **Alpha–beta pruning** skips branches that cannot change the decision.

SEARCH COMPLEXITY

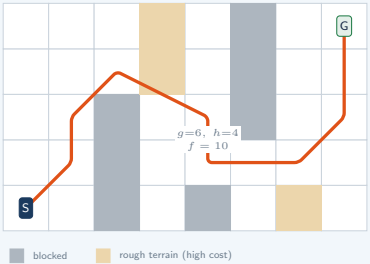
Minimax: $O(b^d)$ $\quad$ α–β pruning: $O\left(b^{d/2}\right)$

A* EVALUATION FUNCTION

$$f(n) = g(n) + h(n)$$

- $g(n)$: cost so far (terrain, fuel, time)
- $h(n)$: heuristic estimate to the goal
- b = branching factor, d = search depth

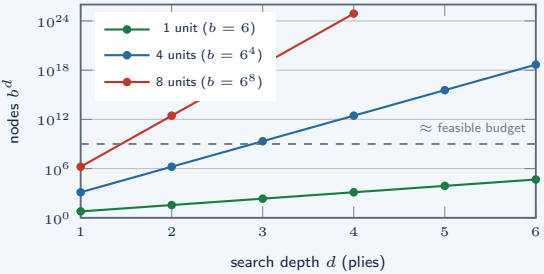
A* route planning on a terrain-cost grid



Wargame numbers

Grid: 50×50 hex, $b = 6$ actions/unit; 8 units $\Rightarrow b_{\text{eff}} \approx 6^8 = 1.68$ million per turn.
Factored per unit (6 actions, depth 4): $6^4 = 1{,}296$ nodes — **feasible**.
Joint search over 8 units: $(6^8)^4 \approx 8 \times 10^{24}$ — **infeasible**.

Nodes to evaluate vs search depth



Why it matters: adding units multiplies the branching factor; depth then makes it exponential. Pruning ($b^{d/2}$) helps, but a modern wargame has $\sim 10^{80}$ states — intractable beyond depth ≈ 8 even with the best search.

Why Classical AI Fails in Real Warfare

Five structural limits ⇒ the case for learning from data

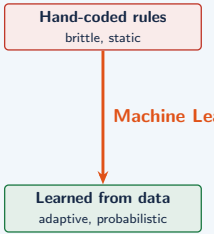
① **BRITTLENESS.** Novel situations = total failure. Gulf War 1991: expert systems failed on **81%** of non-scripted operations.

② **KNOWLEDGE BOTTLENECK.** Encoding all doctrine by hand: the F/A-18 avionics ES needed **10K rules, 200 expert-months**.

③ **NO UNCERTAINTY.** Warfare is stochastic: weather, morale, comms loss are probabilistic events, not Boolean facts.

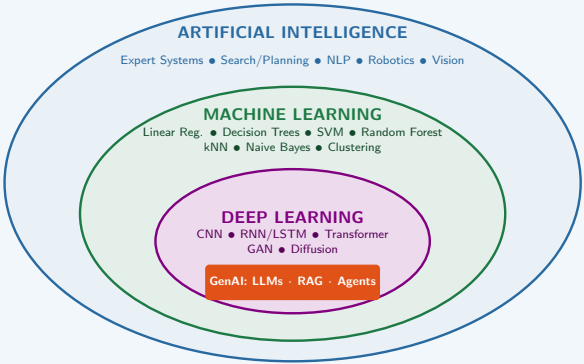
④ **STATE-SPACE EXPLOSION.** Modern wargame $\sim 10^{80}$ states. Even the best search is intractable beyond depth 8.

⑤ **NO LEARNING.** After 1,000 engagements the system is no wiser than at initial deployment.



SOLUTION: let systems **LEARN from DATA**. Machine Learning replaces hand-coded rules with parameters fitted to examples — the rest of this course.

Three Nested Worlds of Intelligent Computing



Artificial Intelligence

Any technique enabling machines to mimic human-level cognition.

Machine Learning

Systems that improve from data without being explicitly programmed.

Deep Learning

Multi-layer neural networks learning hierarchical representations from raw data.

Data dependency scale



Five Core ML Learning Paradigms at a Glance

SUPERVISED	UNSUPERVISED	SEMI-SUPERVISED	SELF-SUPERVISED	REINFORCEMENT
Labelled data: (x, y) pairs Mapping: $X \rightarrow Y$ Task: classification / regression **Defence:** threat ID, force estimation	No labels: $\{x\}$ only Task: clustering, density, anomaly Finds hidden structure **Defence:** tactic clustering, SIGINT grouping	Small D_L + large D_U Propagates labels via graph / EM Mixes supervised + unsupervised **Defence:** limited labelled SAR imagery	Labels derived from the data itself Pretext tasks: mask, rotate, contrast **Defence:** pre-train on raw sensor data Base for LLMs, BERT, MAE	Agent + Environment + Reward Policy: State → Action No labels — learns from interaction **Defence:** autonomous UAVs, wargame agents

Paradigm	Labels?	Data scale	Wargaming example
Supervised	All	1K–10M	Radar threat classification → Fighter / Missile / UAV / Friendly
Unsupervised	None	Any	OPFOR movement archetype discovery (Patrol / Assault / Transit)
Semi-supervised	∼5%	10K+	SAR vehicle recognition with 200 labelled / 20K unlabelled
Self-supervised	None (self)	Large	Signal pre-training on 500K raw radar sweeps (like BERT)
RL	Reward only	Episodes	Blue Force agent training in the JANUS wargame simulation

Learn a mapping $f : X \to Y$ from labelled examples

Mathematical formulation

Training set $D = \{(x_i, y_i)\}_{i=1}^{N}$, with $x_i \in \mathbb{R}^d$ a feature vector and $y_i \in Y$ a label (class or real value). Goal: learn $f : \mathbb{R}^d \to Y$ such that $f(x_{\text{new}}) \approx y_{\text{true}}$.

CLASSIFICATION LOSS (CROSS-ENTROPY)

$$\mathcal{L}(\theta) = -\tfrac{1}{N} \textstyle\sum_i y_i \log f(x_i; \theta)$$

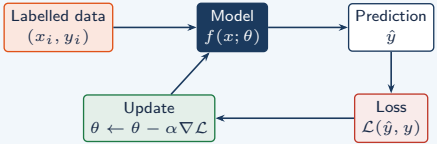
REGRESSION LOSS (MEAN SQUARED ERROR)

$$\mathcal{L}(\theta) = \tfrac{1}{N} \textstyle\sum_i \big(y_i - f(x_i; \theta)\big)^2$$

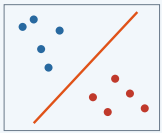
REGULARISED OPTIMISATION & GRADIENT DESCENT

$$\theta^* = \arg\min_{\theta} \big[\mathcal{L}(\theta) + \lambda R(\theta)\big], \qquad \theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}(\theta)$$

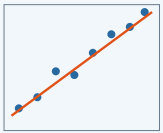
The supervised learning loop



Two main task types



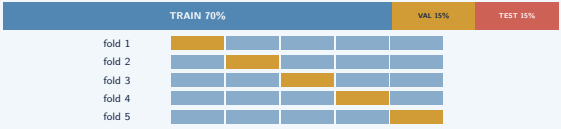
CLASSIFICATION (Y discrete)



REGRESSION ($Y \in \mathbb{R}$)

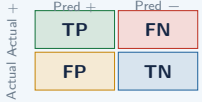
- **Classification:** binary (threat? YES/NO) or multi-class {Fighter, Bomber, Missile, UAV, Civilian}
- **Regression:** casualty estimate, time-to-target of an inbound missile, 72-hour ammo consumption

Evaluation protocol: never touch the test set



5-fold cross-validation: each fold is the validation set once ($k = 5$ or 10)

NEVER use test data during model training! Tune hyper-parameters on validation (or cross-validation) only; touch the test set once, at the end.



Key metrics

CLASSIFICATION

$$\text{Acc} = \frac{TP+TN}{TP+TN+FP+FN} \qquad P = \frac{TP}{TP+FP} \qquad R = \frac{TP}{TP+FN}$$

$$F_1 = \frac{2PR}{P+R}, \qquad \text{ROC-AUC} = \text{area under the ROC curve}$$

REGRESSION

$$\text{MSE} = \frac{1}{N}\sum(y-\hat{y})^2 \qquad \text{RMSE} = \sqrt{\text{MSE}}$$

$$\text{MAE} = \frac{1}{N}\sum|y-\hat{y}|$$

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} \qquad (1 = \text{perfect fit})$$

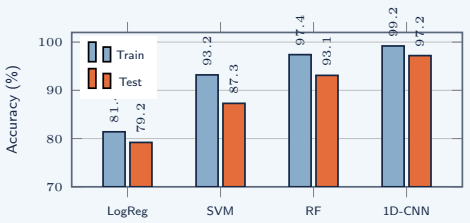
Defence reading: Recall = "of all real threats, how many did we catch?"; Precision = "of all alarms, how many were real?". F_1 blends them but **masks asymmetric costs** — always inspect the confusion matrix.

5-Class Radar Threat Classification for IADS: features and algorithm comparison

Feature	Description	Range	Enemy fighter
x_1 Peak Freq	Radar emission frequency	1–18 GHz	9.35 GHz
x_2 Pulse Width	Duration of radar pulse	0.1–100 μs	2.5 μs
x_3 PRF	Pulse repetition frequency	100–300K Hz	1,500 Hz
x_4 Amplitude	Signal strength at sensor	-120 to -40 dBm	-72 dBm
x_5 Scan Rate	Antenna scan speed	1–60 deg/s	6 deg/s

Dataset and classes

$N = 12{,}500$ samples, 80/20 split. Classes: Friendly Fighter (FF), Adversary Bomber (AB), Missile (MS), UAV (UV), Noise (NZ).



Model	Train	Test	F1	Inference
Logistic Reg.	81.4%	79.2%	0.787	0.2 ms
SVM (RBF)	93.2%	87.3%	0.865	0.8 ms
Random Forest	97.4%	93.1%	0.928	1.2 ms
1D-CNN (DL)	99.2%	**97.2%**	**0.971**	2.1 ms

CNN confusion matrix (test set, $N = 2{,}500$)

Predicted →

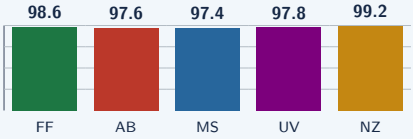
Friendly (500)	FF	AB	MS	UV	NZ
Friendly (500)	**493**	3	1	2	1
Bomber (500)	2	**488**	5	3	2
Missile (500)	3	4	**487**	4	2
UAV (500)	4	2	3	**489**	2
Noise (500)	1	0	1	2	**496**

Rows: actual class (500 each). Red-boxed cell: enemy Bomber labelled Friendly.

Headline result: overall accuracy **97.2%**, macro-F1 **0.971** for the 1D-CNN — the best of four classifiers.

Critical error: threat labelled "Friendly"

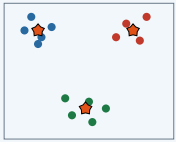
A handful of enemy bombers are classified as Friendly — a **false negative** on the threat class (about 0.6% FN rate on threats as reported). In an IADS, one such miss can be catastrophic.



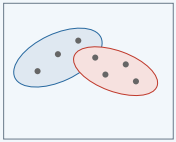
Per-class recall from the matrix (%)

FALSE-NEGATIVE COST: in defence, model choice must optimise **RECALL on the threat class**, not just overall accuracy. F_1 masks this asymmetry; use cost-weighted thresholds and per-class recall targets.

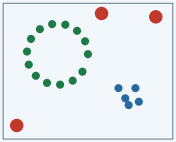
Discover structure in unlabelled data



k-**Means**: assign → recompute centroids



GMM: soft, overlapping, via EM



DBSCAN: arbitrary shapes + noise

The algorithms

k-Means: min $\sum_k \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$; (1) assign x_i to nearest μ_k, (2) recompute $\mu_k = \text{mean}(C_k)$, (3) repeat to convergence.
GMM: $P(x) = \sum_k \pi_k \mathcal{N}(x; \mu_k, \Sigma_k)$; soft assignments, suited to overlapping clusters; fitted by EM (E-step, M-step).
DBSCAN: parameters ε (radius) and MinPts; finds arbitrary shapes; marks low-density points as NOISE (anomalies!).
PCA: reduces d dimensions to $k < d$, retaining the maximum-variance directions; used to compress sensor data before ML.

Defence applications

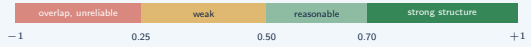
Enemy tactic clustering: group observed OPFOR movements into archetypes without prior labels.
Anomaly detection: network baseline → unusual traffic = likely cyber intrusion.
SIGINT clustering: group intercepts by signal characteristics without prior knowledge of transmission type.
Analyst support: cluster HUMINT reports by topic/actor for prioritisation.

Within-cluster sum of squares (WCSS) vs k



- Big drops until $k = 4$ (Δ253.4), then diminishing returns: Δ19.2 at $k = 5$, Δ7.4 at $k = 6$.
- WCSS *always* falls as k grows — never choose k on WCSS alone.

Validation: silhouette score



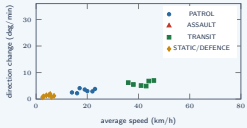
SILHOUETTE

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

- a: mean distance to own cluster; b: mean distance to nearest other cluster
- $s > 0.70$ strong; $s > 0.50$ reasonable; $s < 0.25$ overlapping, unreliable

Decision (case study below): the elbow supports $k = 4$; the silhouette at $k = 4$ is **0.683**, a reasonable-to-strong structure.

Clustering enemy force manoeuvre patterns (JANUS simulator): $N = 4{,}800$ trajectories, 5 features, $k = 4$



Cluster	Size	Speed	DirChg	Spread	Interpretation
0	1142	18.3	3.2	420 m	**PATROL**
1	892	62.7	28.4	85 m	**ASSAULT**
2	670	41.2	6.1	310 m	**TRANSIT**
3	2096	4.8	1.2	380 m	**STATIC / DEFENCE**

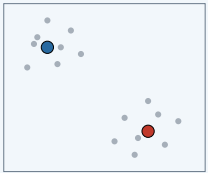
Dataset and validation

Source: JANUS exercise logs. Features (z-score): x_1 speed [0,120] km/h; x_2 direction-change rate [0,45] deg/min; x_3 formation spread [10,500] m; x_4 dwell time [0,60] min; x_5 terrain exposure [0,1]. Elbow $\Rightarrow$ $k = 4$; silhouette **0.683**; Calinski–Harabasz **1,248.7**. Experts labelled 200 random samples: **86.5%** agreement.

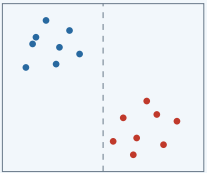
TACTICAL INSIGHT: cluster 1 (ASSAULT) showed 28.4 deg/min direction change — a rapid flanking signature, previously MISSED by rule-based analysts who only monitored a speed threshold > 60 km/h. ML found the pattern *without being told what to look for.*

LIMITATIONS: silhouette $0.683 < 0.75$ (TRANSIT/ASSAULT overlap $\approx 13.5\%$); assumes $k = 4$; doctrine is non-stationary $\Rightarrow$ retrain.

Problem: labelling requires scarce PHOTINT experts, while raw imagery is abundant



Stage 1: 2 labelled + many unlabelled

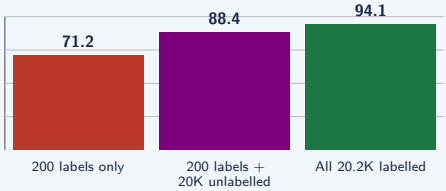


Stage 2: labels propagate along data structure

Defence example: SAR image segmentation

200 expert-labelled SAR images (tanks, APCs, trucks) + 20,000 unlabelled SAR images. **Techniques:** label propagation, pseudo-labelling, consistency regularisation, MixMatch, FixMatch.

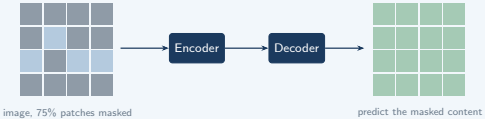
Accuracy by supervision regime



Semi-supervised bridges most of the gap at roughly **12%** of the full-supervision labelling cost: $71.2\% \rightarrow 88.4\%$ vs 94.1% for full labels.

Wargaming: in classified domains, labelled data is scarce and expensive. Semi-/self-supervised methods exploit abundant raw sensor and comms data with minimal expert labelling — crucial for rapid capability development.

Pretext tasks: no human annotation, scales to any data volume



- **Masked autoencoding (MAE):** mask 75% of image patches → predict masked content
- **Contrastive (SimCLR):** two augmented views of the same image must be similar
- **BERT masking:** mask 15% of text tokens → predict them
- Key models: SimCLR, MoCo, DINO, MAE, CLIP

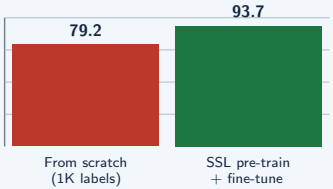
SELF-SUPERVISED PIPELINE

$$\theta^* = \arg\min_{\theta} \mathcal{L}_{\text{pretext}}(\theta, D_U)$$

$\longrightarrow$ fine-tune on D_L $\longrightarrow$ evaluate on held-out test

Defence application: radar signals

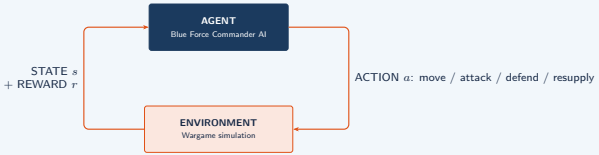
Pre-train on **500K** unlabelled radar signals; fine-tune on only **1,000** labelled threat signals.



+14.5 points from pre-training on unlabelled data alone — significant for defence.

Why it works: filling in a masked patch forces the network to learn the signal's structure; that representation then transfers to the labelled task.

An agent learns an optimal strategy from reward

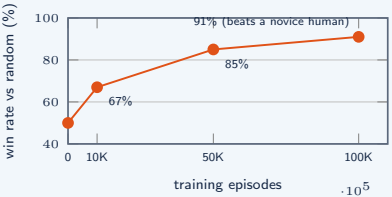


MDP formalism (S, A, P, R, γ)

S state space (what the agent observes); A action space; $P(s'|s,a)$ transition model; $R(s,a,s')$ reward; $\gamma \in [0,1]$ discount (weight of future reward). **Goal:** find $\pi^*(s) = \arg\max_a Q^*(s,a)$ maximising $G_t = \sum_{k\geq 0} \gamma^k R_{t+k+1}$.

Wargame RL example

State: 20×20 hex grid + unit positions + resources. Actions: move (4 dir), attack, defend, resupply $= 7$/unit. Reward: $+10$ objective, $+5$ kill, -8 unit lost, $+100$ win.



No labels, no examples of "the right move": the agent discovers tactics purely from interaction, exactly like a trainee learning from exercises.

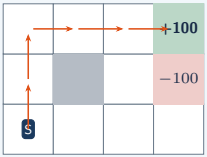
BELLMAN OPTIMALITY EQUATION

$$Q^*(s,a) = \sum_{s'} P(s'|s,a)\Big[R + \gamma \max_{a'} Q^*(s',a')\Big]$$

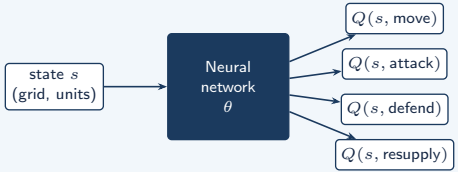
Q-LEARNING UPDATE RULE

$$Q(s,a) \leftarrow Q(s,a) + \alpha\big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big]$$

- α = learning rate; ε-greedy: explore with probability ε, exploit with $1-\varepsilon$
- **Deep Q-Network (DQN):** replace the Q-table with a neural network $Q(s,a;\theta) \approx Q^*(s,a)$ — handles huge state spaces



learned greedy policy π^* (follow arrows)



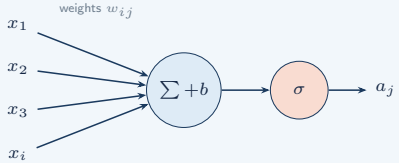
DQN: one forward pass gives the value of every action

Convergence on the wargame (vs random OPFOR)

10K episodes: **67%** win rate; 50K episodes: **85%**; 100K episodes: **91%** (beats a novice human).

Reading the update: the bracket is the TD error — the gap between what we predicted (Q) and what we then observed ($r + \gamma \max Q'$). Each step nudges Q toward the better estimate. Lectures 10–12 develop this in depth.

A single neuron and the network it builds



NEURON, GRADIENT AND UPDATE

$$a_j = \sigma\Big(\sum_i w_{ij} x_i + b_j\Big)$$

$$\frac{\partial \mathcal{L}}{\partial w_{ij}} = \frac{\partial \mathcal{L}}{\partial a_j}\,\sigma'(z_j)\,x_i, \qquad \theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$$

Universal approximation (Hornik 1989)

A network with ≥ 1 hidden layer can approximate any continuous function to arbitrary accuracy — the theoretical foundation of DL.

Activation functions σ



- **ReLU** $\max(0, x)$ — most common
- **Sigmoid** $1/(1 + e^{-x})$ — binary output
- **Softmax** $e^{x_i} / \sum_j e^{x_j}$ — multi-class
- **Tanh** $(e^x - e^{-x})/(e^x + e^{-x})$

SAR / satellite image: what each layer learns

L1 pixel edges; shadow boundaries

L2 corner reflectors; geometric shapes

L3 vehicle silhouettes; track marks

L4 tank vs APC vs truck type

L5 tactical formation; posture

L6+ context; intent inference

more abstract

shallow (1 layer): simple linear boundaries

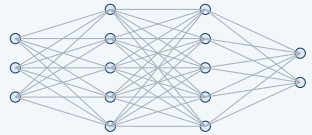
deep (5–100+): hierarchical abstraction

Scale law (Kaplan 2020): more data + more parameters + more compute ⇒ better performance, *predictably*.

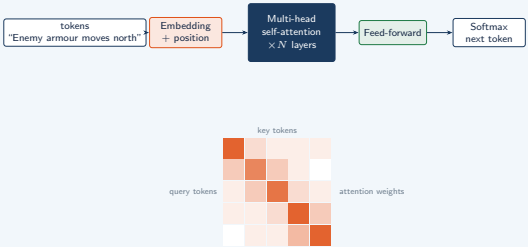
Architectures and defence workhorses

Which architecture for which data?

Net	Data	Result
CNN	Images, SAR, satellite	97.2% target ID
RNN/LSTM	Radar tracks, SIGINT	78% intent prediction
Transformer	Intel reports, LLMs	88% NER on SIGINT
GAN	Synthetic training data	8× data augmentation



LLMs: the Transformer revolution



ATTENTION & PRE-TRAINING OBJECTIVE

$$\text{Attention}(Q,K,V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V$$

$$\max_\theta \; P(w_t \mid w_1,\ldots,w_{t-1};\theta) \quad \text{(next-token prediction)}$$

Architecture and scale

Transformer decoder (autoregressive) with multi-head self-attention. **GPT-3:** 175B parameters, 300B training tokens. **GPT-4:** ∼1.8T parameters (est.), 13T tokens.

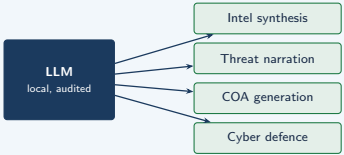
Emergent capabilities at scale

In-context (few-shot) learning • chain-of-thought reasoning • code generation • multi-language.

Fine-tuning options

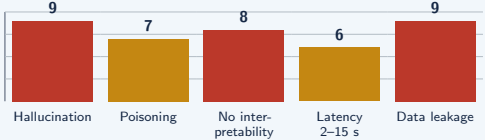
Full: update all θ (expensive). **LoRA:** train small adapter matrices (efficient). **RLHF:** align to human preferences (safety).

Defence applications



- **Intelligence synthesis:** auto-summarise 500+ daily SIGINT reports into a commander-ready brief in < 2 minutes
- **Threat narration:** convert structured sensor output into a natural-language threat assessment
- **COA generation:** suggest courses of action from doctrine + current situation
- **Cyber defence:** analyse malware code, generate YARA rules, explain CVEs to non-technical commanders

Risks in defence



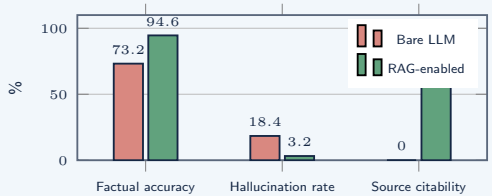
Illustrative qualitative severity for defence use (not measured data)

LLM RISKS IN DEFENCE

- **Hallucination:** confident but WRONG facts — dangerous in battle planning
- **Data poisoning:** adversarially manipulated pre-training data
- **No interpretability:** why did the model recommend this COA?
- **Latency:** GPT-4 API 2–15 s: unacceptable for real-time tactical decisions
- **Classified leakage:** cloud LLMs unsuitable for sensitive operations

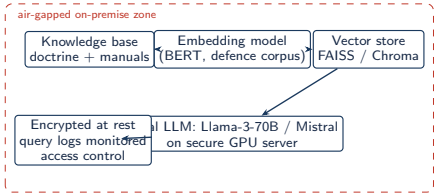
RAG: grounding LLMs in verified, classified knowledge





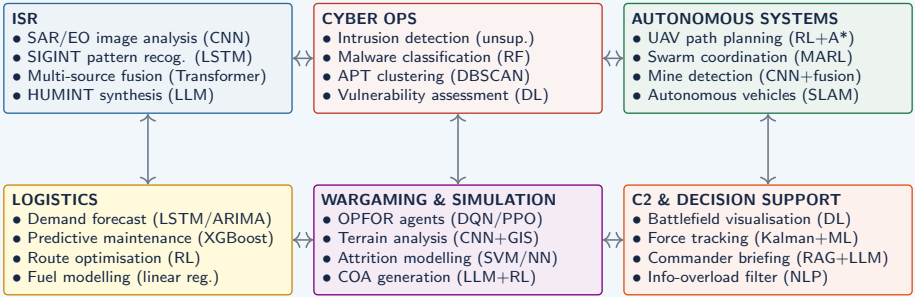
Trade-off: RAG costs about $+1.3$ s latency (2.1 s → 3.4 s) and $+\$0.02$ per query ($\$0.04$ → $\$0.06$) for a **+21.4%** accuracy gain and near-zero hallucination.

Secure defence RAG architecture



No cloud dependency for classified data.

Comprehensive AI/ML application map for defence & wargaming



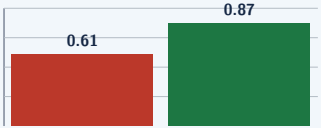
India defence AI: DRDO AI-enabled artillery fire control, ISRO ML-based satellite image analysis, Navy sonar ML systems, Army autonomous surveillance UGVs — active deployments across all six domains. Data flows ISR → C2 → Wargaming → Autonomous Systems in the sensor-to-shooter chain.

Computer Generated Forces (CGF)

Traditional: scripted NPC behaviour — predictable, exploitable → **AI-enhanced OPFOR:** adaptive, doctrine-adherent RL agents

- Multi-echelon: strategic → operational → tactical; adapts mid-exercise to Blue strategy
- DRL OPFOR effectiveness: **78%** of an expert human; cost ∼1/1000th of human OPFOR staffing; available 24/7

Attrition: Lanchester's laws, enhanced by ML

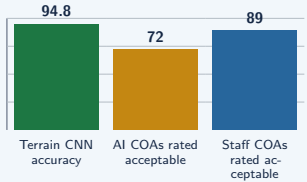


CLASSICAL LANCHESTER

$$\frac{dR}{dt} = -\alpha B, \qquad \frac{dB}{dt} = -\beta R$$

Random Forest evaluated on 1,200 engagements.

Terrain & COA analysis with AI



- **Terrain (ResNet-34):** 50K patches (1 km^2), 5 classes; 94.8%, 11 ms/patch. Manual GIS 4 h per 1000 km^2 vs AI 12 min ($20\times$)
- **COA (LLM+RL):** MDMP planning cycle 6 h vs AI: 3 options in 45 s; quality gap 17 points $\Rightarrow$ initial planning support, not final orders

Wargame AI maturity levels



SIMULATION-TO-REALITY GAP: a 91% win rate in JANUS $\neq$ 91% real-world performance. Sim-to-real transfer, fog of war, adversarial adaptation and friction must be modelled explicitly. Over-confidence in wargame AI results is a known failure mode.

The six critical data pitfalls in defence AI (overview)

1 SURVIVORSHIP BIAS
Study only survivors → ignore failures.
Ex: train only on won battles

2 SPURIOUS CORRELATION
Correlation ≠ causation.
Ex: "lunch quality" predicts exercise wins?

3 OVERFITTING & DIST. SHIFT
Model memorises data; fails in a new environment.
Ex: Afghan terrain → Iraq

4 TRAINING DATA BIAS
Skewed training set; class imbalance; confirmation bias in labels.

5 LABEL NOISE
5% noise → -3.4% acc.; 30% → -35%.
Adversarial poisoning possible.

6 MISUSE OF STATISTICS
P-hacking, cherry-picking; mean without context; Simpson's paradox.

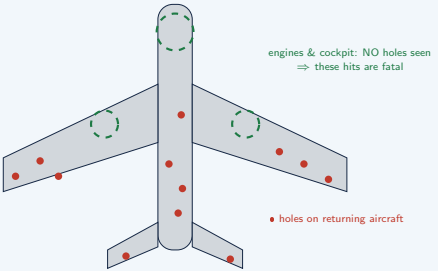
Reference thread

AK Dhamija (SlideShare: Goody Research — Compatibility Mode) presents fundamental flaws in using data and drawing incorrect conclusions from research. The slides that follow extend these ideas with defence examples and AI-specific failures.

Mark Twain: "There are three kinds of lies: lies, damned lies, and statistics." In defence AI the stakes are existential: a mistrained model advising a commander is an operational liability.

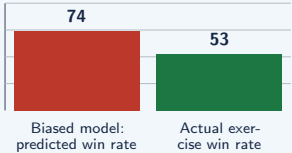
Data Pitfall #1 — Survivorship Bias: Studying Only the Winners

The classic case: Wald & WWII bombers (1943)



- USAF analysed returning B-17s: wings & fuselage most shot up ⇒ "armour those areas!"
- **Wald's correction:** you only see planes that *returned*. Planes hit in the engines/cockpit are at the bottom of the ocean — armour where you see **no** damage.

Wargaming AI application



Training only on battles survived/won predicted **74%** vs an actual **53%**: over-estimation of $+39.6\%$ — catastrophic. **Fix:** seek defeat records, OPFOR after-action reports and classified failure analyses; weight rare failures appropriately.

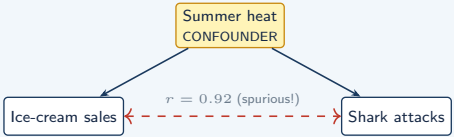
Other military examples

Equipment reliability: failed units are destroyed, so data shows survivors only. Doctrine analysis: operations written up only if successful. Fund rankings: only surviving funds appear.

RED FLAG: if a dataset contains ONLY successes, your model is optimistic by design — ask "Where are the failure records?"

Data Pitfall #2 — Spurious Correlation & Simpson's Paradox

Correlation $\neq$ causation



Defence example (3 years of wargame data)

"Catering quality survey" correlates with exercise win rate, $r = 0.71$. Confounder: budget/readiness. Controlling for budget: $r_{\text{partial}} = 0.09$ ($p = 0.42$) — the effect vanishes.

Simpson's paradox

A trend that appears in every subgroup can **reverse** in the combined data when group sizes are unequal.

Simpson's paradox: always show the counts

	Treatment A	Treatment B
Small stones	81/87 = 93%	234/270 = 87%
Large stones	192/263 = 73%	55/80 = 69%
Combined	273/350 = **78%**	289/350 = **83%**

A wins in *both* subgroups but loses overall: A treated mostly the hard (large-stone) cases. Percentages cannot be added or averaged — totals must come from the counts (see Section 0).

Wargame twin: AI-assisted units 186/300 = 62% vs conventional 256/320 = 80% overall, yet AI units win more in every terrain.

Statistics as a weapon (against you): five key flaws

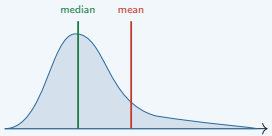
1 MEAN WITHOUT CONTEXT. Average soldier salary Rs 35K/month but median Rs 22K (few senior officers skew the mean) — report median + IQR.

2 P-HACKING. Run 20 tests at $p < 0.05$: expect 1 false positive by chance; $p = 0.048$ means little if 50 other tests were run first.

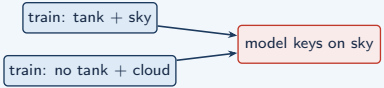
3 CHERRY-PICKING. Reporting only the windows/subgroups that confirm the hypothesis. Full reporting required.

4 SMALL-SAMPLE OVERGENERALISATION. 3 wargame exercises → "AI always outperforms". Need $N \geq 30$ for basic CLT guarantees.

5 ECOLOGICAL FALLACY. Group-level pattern ≠ individual-level: "brigades with more AI tools win more" does not mean a specific brigade will.

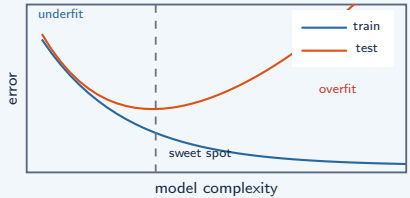


SHORTCUT LEARNING (AI-specific): a CNN classifying tanks learned the *background* (blue sky in the training photos), not the tank. In wargaming: a model infers "night operations" from timestamp metadata rather than thermal signatures — an adversary can exploit this.



Data Pitfall #3 — Overfitting: the Mine Detection Systems

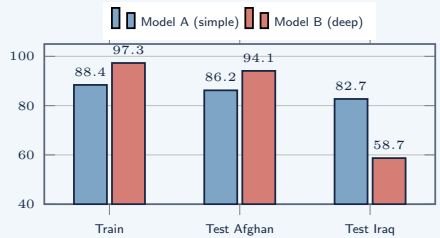
Error vs model complexity



Mine detection (GPR + metal detector)

800 samples, Afghan terrain, 47 signal features. **Model A:** simple CNN, 200K params. **Model B:** deep ResNet, 25M params.

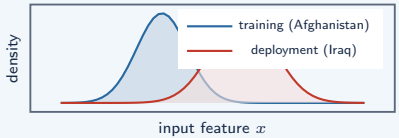
Accuracy by test domain (%)



Model B overfit to Afghan soil signatures: deployed in Iraq it reached only **58.7%** — catastrophic. The bigger, better-looking model was the worse fielded system.

Data Pitfall #3 — Distribution Shift: Detect It Before You Deploy

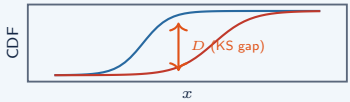
Train vs deployment input distributions



KOLMOGOROV–SMIRNOV SHIFT DETECTOR

$$D = \max_x \left| F_{\text{new}}(x) - F_{\text{train}}(x) \right| \quad \Rightarrow \text{ alert if } D > \tau$$

Causes and remedies

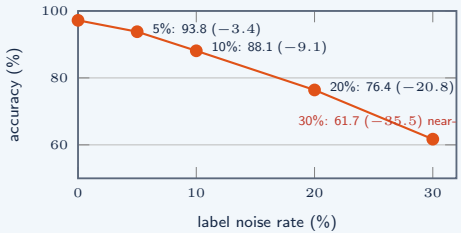


- **Causes:** day vs night imagery; desert vs jungle; 2021 vs 2026 threat profiles
- **Remedies:** L2 regularisation, dropout, early stopping, data augmentation, domain-adaptive training

PROTOCOL: test on data from the **deployment** domain, across all anticipated environments, weather and adversary adaptations, *before* fielding.

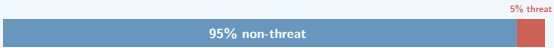
Data Pitfalls #4 & #5 — Training Data Bias and Label Noise

Label noise impact ($N = 10{,}000$ defence sensor samples)



ADVERSARIAL POISONING: 3% poisoning → 94% flip on the target class. Training pipelines must be cryptographically signed and anomaly-monitored.

Training data bias



Class imbalance

Training 95% non-threat / 5% threat → model predicts "non-threat" always: **95% accuracy but ZERO threat detection**. Fix: class weighting, SMOTE oversampling.

Selection bias

Non-representative collection: facial recognition trained on Caucasian faces gave 34% error on darker skin tones (MIT 2018). Intel labels are biased by the analyst's prior beliefs (confirmation bias).

Lecture 1 Summary — Key Takeaways

What we covered today: 15 key points in 6 categories

AI FOUNDATIONS
AI = making machines mimic cognition. Spans classical (rules) to GenAI (learned). Narrow AI is the current state of the art.

ML REVOLUTION
Learn patterns from DATA, not rules. Enabled by data + GPU compute. Explosion post-2012 (ImageNet/AlexNet).

5 PARADIGMS
Supervised / unsupervised / semi-supervised / self-supervised / reinforcement. Each suits a different data-availability scenario.

DEEP LEARNING
Multi-layer NNs: hierarchical features. CNN (images), RNN/LSTM (sequences), Transformer (attention) — the defence workhorses.

GENERATIVE AI
LLMs predict tokens (Transformer decoder). RAG grounds LLMs in verified sources. Defence: ISR synthesis, COA, cyber analysis.

6 DATA PITFALLS
Survivorship bias • spurious correlation • statistical misuse • overfitting/dist. shift • training bias • label noise/poisoning.

Lecture 2 topics

- Decision-boundary concept
- Logistic regression (classification)
- k-Nearest Neighbours (kNN)
- Decision Trees & the CART algorithm
- Random Forest (ensemble methods)
- Support Vector Machines: hard & soft margin, kernel trick (RBF, polynomial)
- Naive Bayes classifier
- Evaluation: precision / recall / F1 / ROC-AUC
- Worked numerical examples throughout

Pre-reading: matrix operations, probability fundamentals (Bayes' theorem) and the sigmoid function — Lectures 2–3 use linear-algebra notation extensively.

Lecture 2 case study (preview)

Multi-class threat classification for IADS: 5 classes (friendly fighter, adversary bomber, cruise missile, UAV/drone, civilian); 11 features (8 radar + 3 EW); $N = 8{,}500$ samples (80/15/5 train/val/test). We will compare **all** algorithms on the same data: which wins? what trade-offs? is 99% test accuracy trustworthy? (*Spoiler: distribution shift will strike again!*) All derivations from first principles, with defence-specific numerical context.

Assignment 1 (released today, Part 3 of this lecture pack): solve the problems at home; bring written solutions to Lecture 2 — the first 5 minutes will discuss common errors.